

Assemble a Subscription Box SOP

One episode assembles one subscription box. The box is already open on its spot with a layout card in the bottom, a tissue sheet under the card hanging over the left and right walls, and a ribbon band threaded under the box. Six numbered items come out of the item tray and go onto their matching outlines on the layout card, the tissue is folded over them and sealed with one sticker, the insert card is laid on top, the lid is closed and its tab tucked, and the two band ends are fastened on the lid.

The order never changes: load, wrap, seal, insert, lid, band. No tissue is folded before all six items are in, no sticker goes down before both tissue folds are made, no insert card goes in before the sticker is pressed, the lid never closes before the card is in, and the band is never fastened before the lid tab is tucked.

The gripper on the item tray's side takes every item out of the tray. The right gripper folds the right tissue overhang and the right lid wing, places the seal sticker, closes the lid, tucks the tab, and fastens the band. The left gripper folds the left tissue overhang and the left lid wing, brings the insert card in, brings the left band end onto the lid, and holds whatever the right gripper is working on. Neither gripper takes the other's zones, nothing is folded or turned in the air, and a gripper opens only over the spot the thing in it is going to.

The table is set up in one of three ways. Only the item tray moves; the box, the sticker pad, the card spot, and the two band ends are in the same place in all three.

- **Config L:** the item tray is at the back-left, behind the left band end.
- **Config M:** the item tray is at the front-center, in front of the box and clear of the near wall.
- **Config R:** the item tray is at the back-right, behind the right band end.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the item tray starts at the back-left (**Config L**), the front-center (**Config M**), or the back-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the item tray's side takes each item — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Fixed roles:** the box, the sticker pad, the card spot, and the band ends never move, and everything after Step 1 — the folds, the seal, the card, the lid, and the band — is done by the same gripper in every config.
- **Order:** load, wrap, seal, insert, lid, band, in every config.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the box at the center, the item tray in the start zone for this episode's config, the sticker pad at the front right, the card spot at the front left, and both ribbon band ends laid out on the table to the left and right of the box.
3. The box is visible from above, so every outline on the layout card, both tissue folds, the sticker press, and the card can be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the box, the item tray, the sticker pad, the insert card, and the two band ends.
6. The left arm reaches the card spot, the left tissue overhang, the left band end, the box, and (Config L and M) every cell of the item tray. The right arm reaches the sticker pad, the right tissue overhang, the right band end, the lid, the box, and (Config R) every cell of the item tray. Neither arm needs the other's zones.

Materials checklist

1. One corrugated **subscription box** sits open and square on the **box spot** at the center of the table. It is a shallow one piece mailer: the **lid** hinges from the far wall, carries a **lid wing** on its left and right edges and a **lid tab** on its free edge, and the **tuck slot** is the gap between the near wall and the inside of the box that the tab drops into.
2. The lid is folded all the way back away from the front edge and lies flat on the table, its printed inside face up, with both lid wings lying flat and open in line with the lid so nothing stands over the box.
3. One **tissue sheet** lies across the bottom of the box, centered, with its left and right long edges hanging over the left and right walls. Each overhang reaches past the middle of the box when it is folded in, and its free edge hangs clear of the table and clear of the band ends.
4. One **layout card** lies flat on the tissue in the bottom of the box, printed face up and square to the box. It carries six numbered **outlines** in two rows of three: outlines 1, 2, and 3 in the far row, left to right, and outlines 4, 5, and 6 in the near row, left to right.

5. One **item tray** sits in the start zone for this episode's config, holding six **items**, one per **cell**. The cells are numbered in the same arrangement as the outlines: cells 1, 2, and 3 in the far row, left to right, and cells 4, 5, and 6 in the near row, left to right. Every item carries a **number label** face up and readable from above.
 - **Config L**: back-left, behind the left band end and clear of the card spot.
 - **Config M**: front-center, in front of the box, clear of the near wall so the lid tab can fold down, and between the card spot and the sticker pad.
 - **Config R**: back-right, behind the right band end and clear of the sticker pad.
6. Each item fits inside its outline with clear space around it and is light enough and small enough for one gripper to lift from above. The box is shallow, so no item stands taller than its walls, and the six items plus the tissue and the insert card sit below the rim with the lid able to lie flat.
7. Which item carries which number varies between episodes. The count stays at six.
8. One **liner square** is taped flat on the **sticker pad** at the front right, carrying one **seal sticker** printed face up. The sticker's near edge is already lifted free of the liner so a gripper takes the lifted edge and the sticker comes away on a straight lift.
9. One **insert card** lies flat on the **card spot** at the front left, printed face up, its long edges running left to right.
10. One **ribbon band** runs under the box, left to right, flat and with no twist. Its **left end** lies flat on the table to the left of the box with its **fastener patch** against the table, and its **right end** lies flat on the table to the right of the box with its fastener patch facing up, so the two patches face each other once both ends are folded over onto the closed lid.
11. The box spot, the table around it, and the space behind the box where the lid lies open are clear and dry.

Workspace layout

- **Box spot**: the center of the table, where the box sits and every load, fold, press, and close happens.
- **Item tray**: the start zone — back-left (Config L), front-center (Config M), or back-right (Config R) — six numbered cells in two rows of three, far row 1, 2, 3 and near row 4, 5, 6, left to right.
- **Sticker pad**: front right, one liner square taped flat with one seal sticker on it, near edge lifted free.
- **Card spot**: front left, one insert card lying flat, printed face up.
- **Band ends**: one on the table to the left of the box, one to the right, both flat.

Apart from the item tray, everything the right gripper works on is staged on the right or at the center, and the insert card and the left band end the left gripper carries are on the left. The item tray is taken by the gripper on its side. No carried thing crosses the midline.

Vocabulary

- **Arm assignment:** the right gripper folds the right tissue overhang and the right lid wing, places the seal sticker, closes the lid, tucks the tab, fastens the band, and in Config R lifts every item out of the item tray. The left gripper folds the left tissue overhang and the left lid wing, brings the insert card in, brings the left band end onto the lid, holds whatever the right gripper is working on, and in Config L and M lifts every item out of the item tray. Neither gripper takes the other's zones and neither takes over the other's work.
- **Box:** the open corrugated mailer on the box spot. Its **near wall** is the wall nearest the front edge and its **far wall** is the one carrying the lid hinge.
- **Lid:** the panel hinged to the far wall. It lies folded back flat on the table, printed inside face up, until Step 5, and closes by swinging over the box on its hinge. It is never lifted off the hinge.
- **Lid wings:** the two flaps on the left and right edges of the lid. They lie flat and open until they are folded up square to the lid, and they drop inside the left and right walls as the lid comes over.
- **Lid tab:** the flap on the free edge of the lid. Once the lid is down, it folds against the outside of the near wall and its edge drops into the **tuck slot**, the gap between the near wall and the inside of the box. A tucked tab is what holds the box closed.
- **Layout card:** the printed card in the bottom of the box carrying the six numbered outlines. It stays in the box for the whole episode and is never lifted.
- **Outline:** one printed shape on the layout card, numbered 1 to 6. Outlines 1 to 3 are the far row and 4 to 6 are the near row, so loading in printed order fills the far row first and the near row never stands in the way of a reach to the far row.
- **Item:** one product going into the box. Its **number label** is read in the item tray and it decides the outline: item N goes into outline N and nowhere else.
- **Start zone:** where the item tray sits at the start of the episode — back-left (**Config L**), front-center (**Config M**), or back-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Tissue sheet:** the sheet under the layout card. Its **left overhang** and **right overhang** are the parts hanging over the left and right walls. The **tissue seam** is the line running front to back along the middle of the box where the two folds overlap.
- **Seal sticker:** the one sticker on the liner square. It is taken by its lifted near edge, carried with the adhesive face down, and pressed once onto the tissue seam.
- **Insert card:** the printed card that goes on top of the sealed tissue, printed face up.
- **Ribbon band:** the band threaded under the box. Its **band ends** are the two lengths lying flat on the table, each carrying a **fastener patch**. The band is fastened by laying the two ends on the middle of the closed lid so the patches meet, and pressing them together.

- **Flat hold:** a gripper closed and resting on a tissue fold, a band end, or a card to keep it from sliding or springing back while the other gripper works. During a fold it lifts clear as the incoming layer reaches it, so the fold lands flat.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower on:** carrying level over the target, lowering straight down until the thing is resting, and releasing once it is resting. Nothing is dropped from height.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing sliding, springing back, or hanging over an edge.
- **Release point:** inside the outline the item is going into, on the tissue seam, on the sealed tissue, on the box rim, or on the middle of the closed lid. A gripper opens nowhere else.

Steps

Steps 1 to 6 assemble the box in one pass, in that order. Step 7 ends the episode once the band is fastened. Step 1 depends on where the item tray is: in Config L and M the **left gripper** loads the items; in Config R the **right gripper** does. Every other line, in Steps 1 to 7, is the same in all three configs.

Step 1: Load the six items onto the layout card

Goal: all six items resting inside their matching outlines, each number label face up.

Look where the item tray is before reaching for the first item.

- **IF the item tray is at the back-left (Config L):** the **left gripper** closes on the item in **cell 1**, lifts it straight up, carries it level forward and to the right over the box, lowers it into **outline 1** until it is resting on the layout card, and releases.
- **IF the item tray is at the front-center (Config M):** the **left gripper** closes on the item in **cell 1**, lifts it straight up, carries it level straight back over the box, lowers it into **outline 1** until it is resting on the layout card, and releases.
- **IF the item tray is at the back-right (Config R):** the **right gripper** closes on the item in **cell 1**, lifts it straight up, carries it level forward and to the left over the box, lowers it into **outline 1** until it is resting on the layout card, and releases.

Then, in all three:

- Do the same for the items in cells 2, 3, 4, 5, and 6, in that order. Item N always goes into outline N.
- Take one item per trip and finish it before the next one comes out of the tray.
- Read the number label in the tray before the lift. It is what picks the outline.
- The other gripper stays clear of the box for the whole step.

Check: all six cells are empty, all six items are resting inside their matching outlines with their number labels face up, no item stands on another, no item rests on a box wall or on the tissue, and no item hangs outside its outline into a neighbouring one. If an item lands outside its outline or on another item, lift it clear and lower it again.

Step 2: Fold the tissue over the items

Goal: both overhangs folded in over the items with the right fold lying on top of the left and the tissue seam running front to back along the middle of the box.

Both folds are made on the box. Never lift the tissue clear of the box and never fold it in the air.

2.1 Fold the left overhang in

- The **left gripper** pinches the middle of the **left overhang** at its free edge, draws it up over the left wall and across the items, lays it down so its free edge lies past the middle of the box, and releases.

Expected state: the left half of the items is covered and the loose edge of the left fold lies past the middle of the box.

2.2 Fold the right overhang over it

- The **left gripper** takes a **flat hold** on the left fold close to the left wall and keeps it while the right overhang comes over.
- The **right gripper** pinches the middle of the **right overhang** at its free edge, draws it up over the right wall and across, lays it down on top of the left fold so the two overlap along the middle of the box, and releases.
- The **left gripper** lifts clear as the incoming edge reaches it.

Check: every item is covered with no item showing through a gap, the right fold lies on top of the left, the two folds overlap along the middle of the box, and no tissue is trapped under an item or hanging outside the box. If a gap is open or a fold pulls back, draw that overhang across again and lay it flat.

Step 3: Seal the tissue with one sticker

Goal: one seal sticker pressed down flat on the tissue seam.

- The **left gripper** takes a **flat hold** on the tissue at the far end of the seam and keeps it for the whole press.
- The **right gripper** closes on the lifted near edge of the **seal sticker**, lifts it straight up off the liner square, carries it level to the box with the adhesive face down, lowers it onto the middle of the **tissue seam**, presses it down for 2 seconds, and lifts straight off.
- The **left gripper** releases the flat hold once the right gripper is clear.

One sticker, one press, on the seam and nowhere else. The sticker is laid down once. Do not lift a stuck sticker to place it again, because the adhesive loses its hold on a second try.

Check: the sticker lies flat across the tissue seam, printed face up, pressed down across its whole width with no corner lifting, and it holds both folds together when the tissue is nudged. If a corner lifts, press along it again.

Step 4: Lay the insert card on the sealed tissue

Goal: the insert card lying flat on the sealed tissue, printed face up and square to the box.

- The **left gripper** closes on the near edge of the **insert card** at the **card spot**, lifts it straight up, carries it level over the box, lowers it flat onto the sealed tissue with its long edges still running left to right, and releases once it is resting.

The card goes down the way it was staged, printed face up. Nothing is turned in the air.

Check: the card lies flat on the sealed tissue, printed face up, fully inside the box with no corner riding a box wall or the lid hinge, and the sticker is still down. If a corner rides a wall, lift the card clear and lower it again.

Step 5: Close the lid and tuck the tab

Goal: the lid lying closed and flat on the box with both wings inside the walls and the lid tab tucked into the tuck slot.

5.1 Fold the two lid wings up

- The **left gripper** pinches the free edge of the **left lid wing** and folds it up until it stands square to the lid, then releases.
- The **right gripper** does the same with the **right lid wing**.

Both wings go up before the lid moves. A wing left lying open catches on the box wall on the way over.

Expected state: the lid still lying back on the table with both wings standing square to it.

5.2 Swing the lid over and tuck the tab

- The **right gripper** closes on the middle of the **lid tab**, draws it toward the front edge so the lid swings over the box on its hinge, and lays the lid down flat on the box so both wings drop inside the left and right walls.
- The **left gripper** takes a **flat hold** on the closed lid near the hinge and keeps it for the tuck.
- The **right gripper** folds the **lid tab** down against the outside of the near wall, presses its edge into the **tuck slot** until the tab is inside and the lid front edge sits down on the near wall, and releases.
- The **left gripper** releases once the right gripper is clear.

The lid closes in one swing, on its hinge. Never lift the lid off the hinge and never press the lid down after the tab is tucked.

Check: the lid lies closed and flat on the box, both wings are inside the walls with neither crumpled or standing outside, the lid tab is fully in the tuck slot, no tissue edge or card corner is trapped between the lid and the rim, and the lid does not spring up when the tab is released. If something is trapped or a wing landed outside a wall, pull the tab back out, swing the lid open, fix it, and close the lid again.

Step 6: Close the ribbon band

Goal: both band ends lying on the middle of the closed lid with the two fastener patches pressed together.

6.1 Bring the left end over

- The **left gripper** closes on the **left band end**, draws it up over the left wall and onto the middle of the closed lid so its **fastener patch** faces up, and takes a **flat hold** on it there. The hold stays on for the whole fastening.

6.2 Bring the right end over and fasten

- The **right gripper** closes on the **right band end**, draws it up over the right wall, and lays it down on the left end so the two fastener patches meet face to face.
- Press the two patches together for 2 seconds, then lift straight off.
- The **left gripper** releases once the right gripper is clear.

The band is fastened once. Do not pull the ends apart and fasten them again.

Check: the band runs under the box and over the lid with both ends meeting on the middle of the lid, the two patches are pressed face to face, the band lies flat along its whole length with no twist, and the band stays fastened when an end is nudged. If the patches missed each other, lift the right end clear, lay it on the left end again, and press.

Step 7: Confirm the assembled box and end the episode

- Confirm the box is closed with the lid flat and its tab tucked, the band fastened on the middle of the lid and flat along its whole length, and confirm the six cells of the item tray are empty, the liner square is bare, the card spot is empty, and the table is otherwise clear.
- Leave the scene untouched for 3 seconds with the box, the item tray, the sticker pad, and the card spot all in frame. Nothing is held or moved during the hold.
- Return both arms home, clear of the box and the item tray, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Pull the two fastener patches apart, lay each band end back flat on the table, and check the band for a twist. It stays threaded under the box.
2. Pull the lid tab out of the tuck slot, swing the lid back open, lay it flat on the table folded back away from the front edge with its printed inside face up, and fold both lid wings back open so they lie flat in line with the lid.
3. Lift the insert card out and lay it flat on the card spot, printed face up, long edges running left to right. Replace it if it is creased or bent at a corner.
4. Take the used seal sticker off the tissue and discard it. Stickers are single use.
5. Open both tissue folds and lay each overhang back over its wall, free edge hanging down the outside and stopping short of the table.
6. Lift the six items out and set each one back in its cell in the item tray, number label face up, and set the tray in the start zone for the next episode's config — back-left (Config L), front-center (Config M), or back-right (Config R).
7. Check the tissue sheet for tears and for a hole where the sticker was. Swap in a fresh sheet if it is torn, holed, or too creased to lie flat, and lay the layout card back on it, printed face up and square to the box.
8. Change which item carries which number label between episodes rather than running the same mapping every time. The count stays at six items and six outlines.
9. Lift the near edge of a fresh seal sticker free of the liner square and leave it on the sticker pad, printed face up. Confirm the liner square is still taped flat.
10. Confirm the box sits square on the box spot, wipe the table around it, and confirm the surface is clean and dry. A dusty table lets a band end slide under a flat hold.
11. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R (item tray on the right side of the table, behind the right band end). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the item tray is not in the start zone for the config; a gripper reaches across the table for an item; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the item tray starts at the back-left, the front-center, or the back-right, and the gripper on its side takes each item — the left gripper in Config L and M, the right gripper in Config R; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the item tray is before the first reach, then follow that config's IF line through Step 1.

Violation: Wrong order

- **Visible cue:** a tissue overhang is folded before all six items are in, the sticker goes down before both folds are made, the insert card goes in before the sticker is pressed, the lid closes before the card is in, or a band end comes onto the lid before the tab is tucked.
- **SOP rule broken:** Steps 1 to 6, the order never changes: load, wrap, seal, insert, lid, band.
- **Coaching note:** load, wrap, seal, insert, lid, band. Finish the step you are in before you start the next one.

Violation: Items loaded out of order

- **Visible cue:** the items leave the tray in an order other than 1, 2, 3, 4, 5, 6, so a near row outline fills before the far row is done and a gripper travels over an item already in the box.
- **SOP rule broken:** Step 1, take the items in printed order 1 to 6 so the far row fills first.
- **Coaching note:** count up from one. The far row goes in before anything blocks it.

Violation: Item in the wrong outline

- **Visible cue:** an item is released on an outline whose number does not match its number label, or it goes down without the label having been read in the tray.
- **SOP rule broken:** Step 1, item N goes into outline N and nowhere else.
- **Coaching note:** read the label in the tray, then find that number in the box before you carry.

Violation: Item left outside its outline or stacked

- **Visible cue:** an item is released overhanging its outline into a neighbouring one, resting on a box wall, sitting on the tissue instead of the layout card, or standing on another item, and

the next item is picked anyway.

- **SOP rule broken:** Step 1, every item rests inside its matching outline, and a bad landing is corrected by lifting it clear and lowering it again.
- **Coaching note:** each item gets its own outline flat on the card. Fix it before the next trip.

Violation: More than one item handled at once

- **Visible cue:** two items are carried in one trip, an item is moved while another is still in the other gripper, or the tray is emptied in a sweep.
- **SOP rule broken:** Step 1, take one item per trip and finish it before the next one comes out.
- **Coaching note:** one item, one outline, one trip.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes an item out of the item tray, folds the right overhang or the right lid wing, places the sticker, closes the lid, tucks the tab, or fastens the band; or the right gripper folds the left overhang or the left lid wing, brings the insert card, or takes the left band end.
- **SOP rule broken:** Steps 1 to 6, the right gripper loads the items, folds the right overhang and the right lid wing, seals, closes the lid, tucks the tab, and fastens the band; the left gripper folds the left overhang and the left lid wing, brings the card and the left band end, and holds what the right gripper works on.
- **Coaching note:** each arm stays on its own side. Nothing crosses the midline.

Violation: Tissue lifted out of the box or folded in the air

- **Visible cue:** the tissue sheet leaves the box during a fold, an overhang is carried across in the air instead of drawn over the wall, or the layout card is lifted out of the box at any point.
- **SOP rule broken:** Step 2, both folds are made on the box, the tissue is never lifted clear of it, and the layout card stays in the box.
- **Coaching note:** draw the overhang over the wall and lay it down. Nothing folds in the air.

Violation: Only one overhang folded, or folded in the wrong order

- **Visible cue:** the sticker goes onto a single fold, the right overhang is folded before the left, or the left fold is laid on top of the right.
- **SOP rule broken:** Step 2, fold the left overhang in first, then lay the right overhang on top of it so the two overlap along the middle.
- **Coaching note:** left first, right on top. The seam is where they overlap.

Violation: Items showing through the wrap

- **Visible cue:** an item is still visible through a gap between the folds, or a fold pulls back off the items after it is released, and the sticker goes down anyway.

- **SOP rule broken:** Step 2, every item is covered with no gap open, and a fold that pulls back is drawn across again and laid flat.
- **Coaching note:** cover everything before you reach for the sticker.

Violation: Flat hold missed or left under the fold

- **Visible cue:** the left gripper never lands on the left fold while the right overhang comes over and the left fold drags along with it, or the gripper is still on the left fold when the right layer lands on top of it.
- **SOP rule broken:** Step 2.2, the left gripper takes a flat hold on the left fold and lifts clear as the incoming edge reaches it.
- **Coaching note:** hold it down through the draw, then get out of the way just before it lands.

Violation: Sticker pressed off the seam

- **Visible cue:** the sticker lands off the tissue seam, on a box wall, on the lid, half off the box, or on an uncovered item, or more than one sticker goes down.
- **SOP rule broken:** Step 3, one sticker pressed onto the middle of the tissue seam and nowhere else.
- **Coaching note:** find the seam first, then bring the sticker down on it.

Violation: Sticker not pressed, or laid down twice

- **Visible cue:** the sticker touches the seam and comes away at once, is pressed at one corner only, lifts at a corner when the tissue is nudged, or is peeled back off the tissue and laid down a second time.
- **SOP rule broken:** Step 3, press the sticker down for 2 seconds, lift straight off, and lay it down once without lifting a stuck sticker to place it again.
- **Coaching note:** down, hold two, straight up. The adhesive will not hold a second try.

Violation: Tissue not steadied during the sticker press

- **Visible cue:** the left gripper is off the tissue while the sticker is pressed, or a fold slides, rucks, or lifts with the sticker on the way out.
- **SOP rule broken:** Step 3, the left gripper takes a flat hold at the far end of the seam and keeps it for the whole press.
- **Coaching note:** the flat hold goes on before the sticker touches and comes off after the gripper is clear.

Violation: Insert card laid wrong

- **Visible cue:** the card lands printed face down, turned across the box, under the tissue instead of on it, riding a box wall or the lid hinge, or on the table beside the box.

- **SOP rule broken:** Step 4, the card lies flat on the sealed tissue, printed face up, long edges running left to right, fully inside the box.
- **Coaching note:** it goes down the way it was staged, flat and square on the tissue.

Violation: Lid wings not folded up, or left outside a wall

- **Visible cue:** the lid is swung over with one or both wings still lying open in line with it, or a wing lands on the outside of a side wall or crumples against the rim and the tuck goes ahead anyway.
- **SOP rule broken:** Step 5.1, fold both lid wings up square to the lid before the lid moves, so they drop inside the left and right walls as it comes over.
- **Coaching note:** both wings up first. A flat wing will not find the inside of the wall.

Violation: Lid tab not tucked

- **Visible cue:** the lid is laid down but the tab is left standing up, lying on the outside of the near wall, or resting on the lid, or the lid springs up when the tab is released, and the band goes on anyway.
- **SOP rule broken:** Step 5.2, fold the lid tab down against the near wall and press its edge into the tuck slot until the tab is inside and the lid front edge sits down on the near wall.
- **Coaching note:** the tab is what holds the box shut. Tuck it before you reach for the band.

Violation: Lid not closed flat, or something trapped under it

- **Visible cue:** the lid stands proud at its free edge, a tissue edge or a card corner is caught between the lid and the rim, an item holds the lid up off the rim, or the lid is pressed down after the tab is tucked.
- **SOP rule broken:** Step 5.2, lay the lid down flat on the box with nothing trapped, and never press the lid down after the tab is tucked.
- **Coaching note:** clear the edges into the box first, then let the lid settle on its own.

Violation: Lid lifted off the hinge

- **Visible cue:** the lid is lifted clear of the box and carried, or it is tucked and then pulled open and closed again without anything being fixed.
- **SOP rule broken:** Step 5.2, the lid closes in one swing on its hinge and is never lifted off it.
- **Coaching note:** draw the tab over. The hinge does the work.

Violation: Band not fastened

- **Visible cue:** an end is left lying on the table, both ends land on the lid but the patches are never pressed together, the patches are pressed back to back instead of face to face, or the band comes apart when an end is nudged.

- **SOP rule broken:** Step 6, lay both band ends on the middle of the closed lid so the patches meet face to face, and press them together for 2 seconds.
- **Coaching note:** patches facing each other, press two, then nudge it to see if it holds.

Violation: Band fastened off the lid or twisted

- **Visible cue:** the ends meet on a box wall, off the edge of the lid, or off to one side instead of the middle of the lid, or the band lies twisted along its length.
- **SOP rule broken:** Step 6, both ends meet on the middle of the closed lid and the band lies flat along its whole length with no twist.
- **Coaching note:** run your eye along the band before you press. Flat all the way, meeting in the middle.

Violation: Band pulled apart and fastened again

- **Visible cue:** the patches are separated after they have taken hold and pressed together a second time, or an end is worked loose and repositioned after the press.
- **SOP rule broken:** Step 6, the band is fastened once and the ends are not pulled apart to fasten them again.
- **Coaching note:** line the end up before it lands, so one press finishes it.

Violation: Band end not held during the fastening

- **Visible cue:** the left gripper is off the left band end while the right end comes over, or the left end springs back off the lid under the press.
- **SOP rule broken:** Step 6, the left gripper takes a flat hold on the left band end on the lid and keeps it for the whole fastening.
- **Coaching note:** anchor the left end first, release it last.

Violation: Turned in the air

- **Visible cue:** an item, the sticker, or the insert card is rotated or flipped between the lift and the set down, so it lands facing a different way than it was carried.
- **SOP rule broken:** Steps 1, 3, and 4, nothing is turned in the air; the orientation it was staged in is the orientation it lands in.
- **Coaching note:** the wrist does not have the range to turn it in transit.

Violation: Dragged instead of lifted

- **Visible cue:** an item, the sticker, or the insert card slides along the tray, the liner, or the table still touching it, with no daylight under it in transit. The tissue draws in Step 2 and the band draws in Step 6 are not this.
- **SOP rule broken:** Steps 1, 3, and 4, lift straight up until there is daylight before moving sideways.

- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Dropped into the box or pressed down

- **Visible cue:** a gripper opens above the box and an item or the card falls in, or an item is pressed or patted down after it is resting on the layout card.
- **SOP rule broken:** Steps 1 and 4, lower the thing until it is resting and release once it is resting.
- **Coaching note:** take it all the way down, then let go. Leave it how it lands.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on an item, the sticker, the card, or a band end without setting it down first.
- **SOP rule broken:** Steps 1 to 6, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the item tray, over the sticker pad, over the card spot, over a box wall, over the closed lid while carrying an item, or over open table.
- **SOP rule broken:** Steps 1 to 6, a gripper opens inside the outline the item is going into, on the tissue seam, on the sealed tissue, on the box rim, or on the middle of the closed lid, and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides an item, the sticker, the card, or a band end into position after releasing it, instead of lifting it clear and placing it again.
- **SOP rule broken:** Steps 1 to 6, correct a bad placement by lifting the thing clear and placing it again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** an item, the sticker, or the card falls to the table or the floor short of its spot, the box is struck and shifts off its spot, the item tray is knocked out of place, an item already in an outline is dislodged, the liner square is torn off the table, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item still in the tray, a tissue overhang still hanging out, the sticker still on the liner, the insert card still on the card spot, the lid open or its tab untucked, a band end still on the table, an arm away from home, an arm still over the box, or recording stopped before 3 seconds of a still scene.
- **SOP rule broken:** Step 7, confirm the closed box, the fastened band, and the empty tray, liner, and card spot; leave the scene untouched for 3 seconds; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm and hold first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective item:** an item that breaks or leaks mid episode, or a number label that cannot be read once it is in its cell. Replace it before the next episode.
- **Item does not fit its outline:** the printed outline is smaller than the item, or two neighbouring outlines are too close for both items to sit clear. Reprint the layout card before the next episode.
- **Defective seal sticker:** a sticker whose near edge will not stay lifted off the liner, that tears on the lift, or whose adhesive will not take hold under a correct press because it has gone dead or picked up lint. Swap in a fresh sticker before the next episode.
- **Liner square lifts** off the table with the sticker because its tape has failed. Tape it flat again before the next episode.
- **Defective tissue sheet:** torn mid episode, or an overhang too short to reach past the middle of the box. Swap in a fresh sheet before the next episode.
- **Box fault:** a lid tab that will not stay in the tuck slot under a correct tuck, a wing or a hinge that has torn at its fold, a lid that has taken a set and will not lie flat, or a box too small to take the layout card flat. Replace the box before the next episode.
- **Band fault:** a fastener patch that will not take hold under a correct press, a band too short for both ends to meet on the lid, or a band that has come out from under the box. Thread it back or swap it before the next episode.
- **Item or band end slides** under a correct flat hold because the table is dusty or slick. Wipe the table before the next episode.

Annotation subtasks (from SOP)

1. Place one item into its matching outline on the layout card

2. Fold the left tissue overhang in over the items
3. Fold the right tissue overhang on top of the left fold
4. Take one seal sticker off the liner square
5. Press the seal sticker onto the tissue seam
6. Lay the insert card on the sealed tissue
7. Fold the two lid wings up
8. Swing the lid over and tuck the lid tab
9. Bring the left band end onto the closed lid
10. Bring the right band end over and fasten the band
11. Return both arms home and end the episode